Production LLM inference workload datasets: a complete catalog

Five production trace datasets and roughly twenty conversation/benchmark datasets exist publicly that can replay real LLM inference traffic patterns. The most valuable are BurstGPT (10.3M traces over 213 days from Azure OpenAI), ACM Digital Library +2 the Azure LLM Inference Traces (per-request timestamps and token counts from Azure production), GitHub Mooncake/Kimi traces (from a Chinese chatbot processing 100B+ tokens daily), USENIX and WildChat-4.8M (the only large dataset combining real conversation text with per-turn timestamps and token counts). ShareGPT V3 remains the universal standard used by vLLM, SGLang, and every major serving framework GitHub — but it lacks timestamps entirely.

The critical insight for building realistic benchmarks: **no single dataset captures everything**. The standard industry approach combines arrival-pattern traces (Azure/BurstGPT) with prompt-distribution datasets (ShareGPT/WildChat) to construct workloads that reflect both temporal dynamics and realistic content.

Production traffic traces with real arrival patterns

These are the crown jewels — actual per-request logs from production LLM serving systems with timestamps, token counts, and arrival patterns that can be directly replayed.

BurstGPT: the largest and longest LLM serving trace

Source: Regional Azure OpenAI GPT service with 3,000+ users arXiv
Access: `github.com/HPMLL/BurstGPT` and HuggingFace `lzzmm/BurstGPT`
Paper: KDD 2025 GitHub (arXiv:2401.17644)
Size: 10.31 million traces over **213 days** Hugging Face +3

Fields per request: `Timestamp` (seconds from epoch), `Session ID` (conversation mode), `Elapsed time` (end-to-end latency), `Model` (ChatGPT/GPT-4), `Request tokens`, `Response tokens`, `Total tokens`, `Log Type` (Conversation/API). GitHub The breakdown is 4.81M ChatGPT API + 0.24M GPT-4 API + 0.16M ChatGPT conversation + 0.08M GPT-4 conversation traces, arXiv with additional extended-collection data.

This dataset captures **real burstiness patterns** Emergent Mind — weekly periodicity for conversation services DOI Emergent Mind (higher weekdays, lower weekends), aperiodic bursts for API services, AI Models arXiv and Gamma-distributed concurrency. ACM Digital Library +2 It includes failure traces (response tokens = 0). arXiv Already integrated into vLLM as a first-class benchmark dataset (`--dataset-name burstgpt`). vLLM Does not contain actual conversation text (privacy-preserving). DOI ACM Digital Library License is open-source for research.

Suitability: Excellent. Best available for long-term workload characterization, arrival pattern modeling, and load replay.

Azure LLM Inference Traces (2023, 2024, 2025)

Source: Microsoft Azure production LLM inference services GitHub
Access: `github.com/Azure/AzurePublicDataset`
License: CC-BY Attribution GitHub

Three progressively richer releases exist:

- **AzureLLMInferenceDataset2023** — Single day (November 11, 2023) from two services: GitHub code generation and conversation. GitHub +2 Fields: `TIMESTAMP` , `ContextTokens` , `GeneratedTokens` . The coding service had **median prompt size of 1,500 tokens**; conversation service had median of 1,020 tokens. Microsoft Published alongside the Splitwise paper (ISCA 2024). GitHub GitHub Zenodo DOI: 10.5281/zenodo.11003049. GitHub Zenodo

- **AzureLLMInferenceDataset2024** — One full week GitHub (May 10–19, 2024) from two services. Same fields. Published alongside DynamoLLM (HPCA 2025). GitHub GitHub **Much better for capturing diurnal and weekly patterns** than the 2023 single-day snapshot.

- **AzureLMMInferenceDataset2025** — One week GitHub (October 15–22, 2024) from a multimodal inference cluster. GitHub Fields: `TIMESTAMP` , number of images, `InputTokens` , `OutputTokens` . Published alongside ModServe (SoCC 2025). GitHub GitHub **First production multimodal inference trace** with per-request image counts.

These traces are the **most widely cited in LLM serving papers** — used by Splitwise, DynamoLLM, Vidur, SageServe, and many others. ResearchGate The Vidur simulator (`github.com/microsoft/vidur`) includes pre-processed versions at `./data/processed_traces/splitwise_conv.csv` and `splitwise_code.csv` .

Suitability: Excellent. Gold-standard production traces. The main limitation is the relatively short collection periods (1 day to 1 week).

Mooncake/Kimi traces: real Chinese LLM chatbot production

Source: Kimi chatbot by Moonshot AI USENIX (processes 100B+ tokens daily across thousands of nodes) USENIX ACM Digital Library
Access: `github.com/kvcache-ai/Mooncake/tree/main/FAST25-release/traces`
Paper: FAST 2025 **Best Paper Award** GitHub +2 (arXiv:2407.00079)

JSONL format traces with request-level timestamps, input/output token lengths, and prefix-sharing information. Initially released July 2024, updated February 2025 for the FAST'25 paper. GitHub Particularly valuable for **long-context scenarios** — Kimi is known for its 200K+ context window Wikipedia and long-document processing workloads.

Suitability: Excellent. Highly complementary to Azure traces — represents a Chinese production chatbot service with different usage patterns, especially around long-context workloads.

Conversation datasets with realistic prompt distributions

These datasets provide the actual text content that production traces lack, enabling construction of workloads with realistic token distributions, multi-turn patterns, and topic diversity.

WildChat-4.8M: the most complete conversation dataset for benchmarking

Source: Custom HuggingFace chatbot offering free GPT-3.5-Turbo and GPT-4 access
Access: HuggingFace `allenai/WildChat-4.8M` (3.2M non-toxic) and `allenai/WildChat-4.8M-Full` (4.74M, gated)
Paper: ICLR 2024 ("WildChat: 1M ChatGPT Interaction Logs in the Wild")
License: ODC-BY

WildChat is uniquely rich. Every conversation turn includes: `content`, `role`, per-turn `language` detection, `toxic` flag, `redacted` PII flag, **`token_counter`**, **Unix `created` timestamp**, `turn_identifier`, `temperature`, `top_p`, `system_fingerprint`, and a `usage` struct with **`prompt_tokens`, `completion_tokens`, `total_tokens`**. Hugging Face Conversation-level metadata includes `model` (GPT-3.5-turbo, GPT-4, o1-preview, o1-mini), `hashed_ip`, `country`, `state`, and `accept-language` headers. Hugging Face

The 4.8M version contains **111,836 reasoning model conversations** (o1-preview, o1-mini). Hugging Face Multiple scale variants exist: WildChat (650K), WildChat-1M (1M), WildChat-4.8M (4.8M).

Suitability: The best available dataset for replay benchmarking that includes both text

and metadata. Per-turn timestamps enable arrival pattern reconstruction. Per-turn token counts enable precise input/output length distribution modeling. The `usage` struct mirrors the OpenAI API response format exactly.

LMSYS-Chat-1M: the canonical conversation dataset

Source: Vicuna demo and Chatbot Arena (chat.lmsys.org), April–August 2023 Hugging Face
Access: HuggingFace `lmsys/lmsys-chat-1m` (gated)
Paper: arXiv:2309.11998
Size: 1,000,000 conversations Hugging Face across 25 LLMs, 210K unique IPs, Analytics Vidhya +2 154 languages arXiv

Fields: `conversation_id`, `model`, `conversation` (OpenAI JSON format turns), `turn`, `language`, `openai_moderation` flags. Hugging Face Average turns per conversation: **2.0**. Average tokens per prompt: **69.5**. Average tokens per response: **214.5**. Emergent Mind No timestamps in the public release. Token counts must be computed from text.

License: Custom LMSYS license (gated, allows research + commercial use, prohibits redistribution). Hugging Face

Suitability: Moderate. Large and diverse but lacks timestamps and token count metadata. Excellent for input/output length distribution analysis and model diversity, but cannot reconstruct temporal patterns.

ShareGPT V3: the universal benchmark standard

Source: ShareGPT.com (Chrome extension for sharing ChatGPT conversations)
Access: `huggingface.co/datasets/anon8231489123/ShareGPT_Vicuna_unfiltered` — specifically `ShareGPT_V3_unfiltered_cleaned_split.json` (604MB)
Size: ~53K conversations after cleaning Hugging Face Hugging Face
License: Apache-2.0

Fields per conversation: `id`, `conversations` (list of `{from: "human"/"gpt", value: text}`). No timestamps, no token counts, no model metadata. Multiple cleaned variants exist (English-only, no-moralizing, etc.).

This is **THE standard benchmark dataset** across the entire LLM serving ecosystem. vLLM hardcodes the download URL. GitHub vLLM SGLang auto-downloads it when `--dataset-path` is blank. Sglang LMDeploy uses it as the primary benchmark input. Readthedocs The vLLM benchmark suite extracts the first human/GPT turn pair from each conversation to construct prompt/output pairs. vLLM Forums vLLM vLLM's v0.6.0 blog states they use "500 prompts randomly sampled from ShareGPT dataset with fixed random seed" as a standard workload. vLLM

Suitability: Moderate for realism, but indispensable in practice. Its universal adoption makes it essential for cross-framework comparison, despite lacking temporal metadata.

Chatbot Arena Conversations (with timestamps)

Access: HuggingFace `lmsys/chatbot_arena_conversations`
Size: 33,000 conversations LMSYS Org from 20 LLMs, April–June 2023 Hugging Face
License: CC-BY-4.0 (prompts), CC-BY-NC-4.0 (outputs) Hugging Face

Notably includes **timestamps**, pairwise model comparisons (model_a, model_b), user preference votes, and full conversation text for both models. Hugging Face Smaller scale limits utility for large-scale replay.

Agentic and tool-calling trace datasets

These datasets capture the growing category of agent-style workloads with tool calls, multi-step reasoning, and expanding context per turn.

Berkeley Function Calling Leaderboard (BFCL)

Access: HuggingFace `gorilla-llm/Berkeley-Function-Calling-Leaderboard`
Size: 2,000 question-function-answer pairs
Categories: Simple, multiple, parallel, and parallel-multiple function calls; relevance detection; Java/JavaScript/REST API tests; live API tests
Evaluation: AST-based and exec-based metrics
Paper: arXiv:2402.15491
URL: gorilla.cs.berkeley.edu/leaderboard.html

SWE-bench and agent execution trajectories

SWE-bench itself (`princeton-nlp/SWE-bench` on HuggingFace/GitHub) defines 2,294 real GitHub issue resolution tasks. Hugging Face More valuable for inference benchmarking are the **published agent trajectories**:

- **SWE-smith Trajectories** — HuggingFace `SWE-bench/SWE-smith-trajectories` : **5,017 trajectories** from Claude 3.7 Sonnet running SWE-agent Hugging Face
- **Nebius SWE-agent Trajectories** — `nebius/SWE-agent-trajectories` : **80,036 trajectories** from various models; Nebius `nebius/SWE-rebench-openhands-trajectories` : **67,074 trajectories** (Qwen3-Coder-480B + OpenHands v0.54.0, 32,161

successful). Nebius Licensed CC-BY-4.0.

- **SWE-bench experiments repository** (`github.com/SWE-bench/experiments`) contains execution logs and predictions on a public S3 bucket. GitHub

These trajectories capture **real multi-step agentic workflows with growing context per turn** — exactly the pattern seen in Cursor, Devin, and similar coding agents.

ToolBench/ToolLLM and API-Bank

ToolBench (Tsinghua University) provides instruction-tuning data based on **16,000 APIs from RapidAPI** with three complexity levels: arXiv G1 (single-tool), G2 (intra-category multi-tool), G3 (intra-collection multi-tool). Internet Archive +2 **API-Bank** (Alibaba DAMO, HuggingFace `liminghao1630/API-Bank` , EMNLP 2023) contains 314 tool-use dialogues with 753 API calls across 73 tools arXiv and 1,000 domains. Hugging Face GitHub

Other notable agentic datasets

- **AgentBench** (THUDM, ICLR 2024): 8 environments including OS, database, knowledge graph, web browsing OpenReview — `github.com/THUDM/AgentBench`
- **NexusRaven** (Nexusflow.ai): 9 real-world API tasks on HuggingFace `Nexusflow/NexusRaven_API_evaluation` , CC-BY-NC-4.0 Hugging Face
- **Salesforce xLAM**: HuggingFace `Salesforce/xlam-function-calling-60k` — 60K verified function-calling samples Analytics Vidhya

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Coding assistant datasets and interaction traces

DevGPT: real developer-LLM interactions linked to GitHub

Access: `github.com/NAIST-SE/DevGPT` and Zenodo
Size: 29,778 prompts with ChatGPT responses including 19,106 code snippets; arXiv 4,733 shared ChatGPT links from 3,559 GitHub/Hacker News references arXiv
Period: July–October 2023 arXiv
Languages: Python (6,084), JavaScript (4,802), Bash (4,332) as top 3 arXiv
Unique feature: Links conversations to actual source code commits, issues, PRs, and discussions ACM Digital Library arXiv

CodeChat: coding conversations extracted from WildChat

Access: HuggingFace `Suzhen/CodeChat`

Size: 82,845 developer-LLM conversations with 368,506 code snippets in 20+ languages Hugging Face

Key finding: Median token-length ratio of 14:1 (response:prompt); 68% are multi-turn conversations. Hugging Face This ratio is critical for inference workload modeling — coding assistant responses are dramatically longer than prompts.

Repository-level code completion benchmarks

- **CrossCodeEval** (Amazon Science, NeurIPS 2023): ~10K examples from ~1K repos in Python/Java/TypeScript/C#, specifically requiring cross-file context. arXiv `github.com/amazon-science/cceval`
- **RepoBench** (UCSD, ICLR 2024): 10,345 Python + 14,956 Java repositories with retrieval, completion, and pipeline tasks. ResearchGate HuggingFace `tianyang/repobench_python_v1.1` . `github.com/Leolty/repobench`
- **The Stack v2**: 67.5TB in 600+ languages (BigCode project) — the foundation for StarCoder2's FIM training Hugging Face

GitHub Copilot and Cursor have not published raw interaction traces. Copilot provides aggregate organizational metrics (DAU, acceptance rates, lines suggested). Cursor provides team analytics dashboards. The CursorCore paper (arXiv:2410.07002) describes their data pipeline but doesn't release traces.

RAG, long-context, and retrieval datasets

No standardized public dataset of enterprise RAG production traces exists. This is a significant gap. The closest approximations are evaluation benchmarks that provide realistic input length distributions.

Long-context evaluation benchmarks with realistic distributions

- **LongBench** (THUDM): Multi-task long-context benchmark across 6 task categories. Hugging Face HuggingFace `THUDM/LongBench` . Widely used by MLPerf v5.0+ for Llama 3.1 405B evaluation alongside RULER and GovReport. NVIDIA
- **RULER**: Extends needle-in-a-haystack with retrieval, multi-hop tracing, aggregation, and QA at varying complexity levels. Evidently AI arXiv Used by MLPerf for long-context evaluation.
- **InfiniteBench**: Very long context tasks pushing beyond 100K tokens. arXiv
- **Needle-in-a-Haystack variants**: The original (Kamradt, 2023), GitHub Evidently AI

NeedleBench (OpenCompass), Readthedocs Sequential-NIAH (14,000 samples, 8K–128K tokens), arXiv NoLiMa (latent association retrieval), ICML DENIAHL, arXiv BABILong.

Retrieval/RAG source datasets

- **Natural Questions** (Google): 323K real Google Search queries with Wikipedia passage answers Milvus
- **MS MARCO** (Microsoft): 1M+ real Bing search queries with 8.8M passages and relevance labels Artech-digital
- **MTEB** (Massive Text Embedding Benchmark): 56+ datasets across 8 task types, 112+ languages Medium — the standard for evaluating embedding models used in RAG Unstructured
- **MultiDoc2Dial** (IBM): 4,500+ dialogues grounded in 450+ documents across 4 domains ResearchGate — directly tests RAG-style multi-turn, multi-document grounding Academia.edu

Enterprise-specific: **EnronQA** (103,638 emails, 528,304 QA pairs) ResearchGate and **WixQA** (Wix.com customer support, 200 expert + 6,222 synthetic QA pairs) exist but are limited in scale.

What every major serving framework actually benchmarks with

The universal standard across the ecosystem is **ShareGPT V3** (`ShareGPT_V3_unfiltered_cleaned_split.json`). Here is exactly what each framework uses:

Framework	Primary dataset	Also supports	Arrival pattern source
vLLM	ShareGPT (first-class), Sonnet, BurstGPT	Any HF dataset, random, custom JSONL, MT-Bench, vision/audio datasets	Poisson or BurstGPT timestamps
SGLang	ShareGPT (auto-downloads)	Random, shared-prefix, image, Mooncake trace format	Configurable, Mooncake trace replay
LMDeploy	ShareGPT	Synthetic with configurable	Configurable

		tokens	concurrency
TensorRT-LLM	Synthetic (normal/uniform distribution)	CNN/DailyMail, any HF dataset	Configurable
TGI	Synthetic, OpenOrca	Via vLLM's benchmark script	k6 load test scripts
GenAI-Perf/AIPerf	Synthetic, OpenOrca, CNN/DailyMail	ShareGPT, Mooncake payload format	Configurable
MLPerf Inference	OpenOrca (70B), CNN/DailyMail (8B), LongBench+RULER (405B)	GSM8K, MBXP (Mixtral)	Poisson (server scenario)
InferenceX	Random/synthetic (currently)	Plans to adopt ShareGPT	Concurrency sweeps

vLLM's `benchmark_serving.py` supports backends for `vllm`, `sglang`, `tgi`, `lmdeploy`, `deepspeed-mii`, `openai`, `tensorrt-llm`, and `scalellm` Medium GitHub — making it the de facto cross-framework benchmarking tool. The **Sonnet dataset** (`benchmarks/sonnet.txt`) is a text file of Shakespeare's Sonnets included in the vLLM repo, used for controlled prefix-caching benchmarks with configurable input (550), output (150), and prefix (200) token lengths. vLLM

Multi-modal and specialized traces

- **Azure LMM Inference 2025** (`AzureLMMInferenceDataset2025`): The only production multimodal trace — one week from an Azure LMM cluster with per-request `TIMESTAMP`, image count, `InputTokens`, `OutputTokens`. GitHub CC-BY license. GitHub GitHub
- **ShareGPT4V**: 1.2M high-quality image-text pairs (100K GPT-4V captions + 1.24M Share-Captioner captions). HuggingFace `Lin-Chen/ShareGPT4V`. CC-BY-NC-4.0. Hugging Face
- **Video-MME** (CVPR 2025): 900 videos, 2,700 QA pairs, durations from 11 seconds to 1 hour. Emergent Mind Used by GPT-4.1 and Gemini 2.5 Pro as an "industry standard" multimodal benchmark. GitHub
- **Speech/Audio**: LibriSpeech remains the ASR standard (used by MLPerf v5.1). MLCommons MLCommons vLLM supports audio datasets including

`openslr/librispeech_asr`, `facebook/voxpopuli`, and `speechcolab/gigaspeech`.
vLLM

No standardized production VLM or speech inference trace datasets with real arrival patterns exist publicly.

Alibaba and other infrastructure traces worth knowing

Alibaba has released several cluster traces, but none are LLM text inference request-level traces:

- **GenTD26** (`cluster-trace-v2026-GenAI`): Production Stable Diffusion serving traces with 3-layer coverage (application, middleware, infrastructure). GitHub GitHub Has request-level timestamps at the application layer but covers image generation, not LLM text.
- **GPU Cluster v2025**: 23,871 DLRM inference instances — disaggregated recommendation model serving, not LLM. GitHub
- **GPU Cluster v2020/v2023**: Job-level training/inference scheduling traces from PAI platform. GitHub

SageServe (Microsoft, arXiv:2502.14617) describes week-long traces from **Microsoft 365 Copilot** across multiple US data centers with 10M+ requests, 3 regions, and 4 LLMs arXiv — claims open-source artifact release but availability is unconfirmed.

ByteDance, Google, Tencent: No publicly available production LLM inference request-level traces found. Google Borg traces are general-purpose compute GitHub (pre-LLM era). Helios (SenseTime) arXiv and Philly (Microsoft) traces capture DNN training job scheduling, GitHub not per-inference requests.

Conclusion: building realistic benchmark workloads

For building benchmark workloads that reflect what companies like Cursor, Harvey, Glean, and Perplexity see in production, **five datasets stand above the rest**:

1. **BurstGPT** for long-term arrival patterns and burstiness modeling arXiv alphaXiv — 213 days of real traffic ACM Digital Library arXiv with per-request timestamps, Hugging Face +2 already integrated in vLLM vLLM
2. **Azure LLM Inference 2024** for authoritative production traffic from the world's largest

cloud provider — one-week code and conversation traces with per-request timestamps and token counts GitHub GitHub

3. **WildChat-4.8M** for the unique combination of real conversation text, per-turn timestamps, per-turn token counts, and massive scale — the only dataset enabling both content-aware and timing-aware workload construction

4. **Mooncake/Kimi traces** for long-context chatbot production patterns from a service processing 100B+ tokens daily USENIX ACM Digital Library

5. **ShareGPT V3** as the universal cross-framework standard for prompt distribution benchmarking

The standard industry approach in LLM serving papers combines Azure/BurstGPT arrival traces with ShareGPT/WildChat prompt distributions. For agentic workloads, Nebius's 80K+ SWE-agent trajectories Hugging Face and the BFCL dataset provide the closest approximations to real tool-calling patterns. The largest gap remains **enterprise RAG production traces** — no public dataset captures real RAG query logs with retrieval latencies, chunk counts, and generation patterns. The second major gap is **IDE coding assistant traces** — neither Copilot nor Cursor has released interaction-level data.